Volume III · Chapter 15 — Evaluation, Bias, Fairness & Clinical Safety · Theory

> Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-III-Chapter-15-context.md` for the AI interaction log.

Prerequisites: Ch 10, Ch 14.

Learning outcomes — the student can: design rigorous clinical evaluation; measure subgroup bias/fairness; assess calibration and safety (red-teaming); explain the benchmark-to-clinic gap; run clinician-in-the-loop evaluation.

Topics (theory) — 12 h:

#	Topic	h
15.1	Evaluation beyond accuracy: task-appropriate metrics	2
15.2	Bias & fairness: subgroup analysis, disparate performance	2
15.3	Calibration & uncertainty in clinical models	2
15.4	Safety: harmful outputs, refusal, red-teaming	2
15.5	Benchmark vs. clinical validity; prospective vs. retrospective	2
15.6	Human evaluation & clinician-in-the-loop protocols	2

Datasets/tools: `evaluate`, fairness libraries; local LLM. **Assessment:** safety/fairness eval report (**60%**); red-team catalog (**20%**); quiz (**20%**). **Key decisions:** which subgroups & metrics; acceptable risk thresholds; when a model is **not** safe to deploy. **References:** plan §13 → *Medical LLMs*; *Fine-tuning, alignment & benchmarks*. **Hours:** Theory **12** + Lab **10** = **22**.